

Assignment 2 - AI Art & Creative Tools Pals

Overview

For this assessment task, you will design, implement, and debug a full-stack web application, creating both a back-end API and appropriate front-end.

Important: This assignment specification is generated just for you, 30438140. Do not distribute this specification. If your personal specification is made publicly available online, academic misconduct charges may be applied.

Validation code 275a1

Timelines and Expectations

Percentage value of task: 30%

Due: Refer to Unit Description

Learning Outcomes Assessed

The following course learning outcomes are assessed by completing this assessment:

- **K2.** Contrast the capabilities and limitations of client-side and server-side web code
- **K3.** Detect opportunities for increasing security and privacy of web applications
- **S1.** Develop client/server web applications using client-side and server-side code
- **S2.** Connect to and manipulate a database management system programmatically using server-side code
- **S3.** Design and implement a web-based Application Programming Interface (API)
- **A1.** Design, develop, test, and debug client/server web applications to provided specifications

Assessment Details

For this assignment, you will build upon skills acquired through the course and completion of prior assignment work to develop **AI Art & Creative Tools Pals**, an online **link sharing site** for AI art & creative tools enthusiasts to gather and make friends.

The AI art & creative tools link sharing site has the following **functional requirements**:

- **Visitors** (i.e. members of the public) can sign up to become a **Member**, supplying a username and password;
- Once signed up, members can add **Links** that they have found, with a title and a description;
- Both visitors and members can see a global list of **Links** (the “*front page*”);
- Members can give a **rating** to each link - it is up to you to decide what form this rating takes. A single member can only rate any given link once;
- The front page must show some kind of aggregate rating for each link, and should be sortable by *Most Recent* or *Highest Rated*;
- Members can view a list of links they have given a positive rating (ie *favourites*);
- Members **gain Tools Points** if people rate their links positively, and **lose Tools Points** if people rate their links negatively; and
- Members can choose to **hide** particular links, without affecting the rating.

AI Art & Creative Tools Pals has the following **non-functional requirements**:

- You must use Deno + HTML and Vanilla Javascript. **The use of any other framework will be rewarded 0 marks**
- The site must be structured as an HTML front-end with a RESTful API back-end; Compared to your previous assignment, it's ok to have multiple HTML pages, and a mix of dynamically-generated HTML and API-based functionality;
- Apply some very basic visual design that will be attractive to AI art & creative tools enthusiasts;
- Links in the front page or favourites page should display with the name and Tools Points of the member who submitted it;
- At minimum, the following functions, if implemented, must use a RESTful API approach, without whole-page reloads at the client:
 - Posting a new link
 - Rating a link
 - Hiding a link
- The API must be protected from unauthorized use - that is, members should not be able to rate links on behalf of other users via the API, for example;
- Links, ratings and member information must be saved in a relational database; and
- Passwords must be securely stored (use bcrypt at minimum, not plain text or MD5)

Database

Include enough initial data to be an effective demonstration of your site - at least 3 users, 5 links, and 10 ratings. At least one link should have no ratings at all. You must use PostgreSQL (as covered in the unit materials) as your database backend. The use of any other database will result in a loss of 10 marks.

- **Include an SQL script** that will generate your database with your assignment submission. username must be itech3108, password is itech3108pass. Such a script should also create the database. Please name your database **ITECH3108_30438140_a2**. **Include this data as part of your written report, including a usable password for at least one user.**

(It's ok if multiple users have the same password)

Report

Include in your submission a report containing:

- A brief description of your assignment architecture, including a list of URL patterns used in your API;
- A statement indicating which features of the assignment you completed;
- A statement indicating what help you received, if any, from outside sources. Include this statement even if you completed the assignment entirely on your own. You do not need to acknowledge ITECH3108 course materials.

If you include the creative works of others in your database (existing text), acknowledge those sources appropriately.

Hints

This assignment requires you to demonstrate **complex problem-solving skills**; minimal guidance on the structure of your solution will be provided in this assignment specification.

Your database design, API design, back-end and front-end software architecture are entirely up to you, supported by lecture and lab activities. You can design how Tools Points work. Enjoy and/or fear the freedom!

You are free to discuss the architecture of your solution with your peers, as long as you acknowledge them

in your report. Under no circumstances share code solutions.

Bonus optional fun challenge activity 1 - Better Average Ratings

Sorting by ratings is often done using an *average rating*. Is a link with just one positive review and no negatives better than one with 50 positive reviews but one negative? Using an average rating suggests yes!

Bonus optional task: Solve the aforementioned problem so that the ratings take into account the uncertainty around review scores. You might consider so-called **Bayesian Average Ratings**. Explain your approach in the report.

There are insufficient marks allocated to this task, it's not based on any course materials, and is just something exciting you can try.

It is possible to attain full marks for this assignment without completing this challenge task.

Bonus optional fun challenge activity 2 - Svelte, Vue, or React

There are many, many ways to write front-end web applications. Explore one!

Bonus optional task: In addition to your regular ("vanilla") JS front-end, reimplement your front end using one of these three popular front-end frameworks:

- Svelte - <https://svelte.dev/>
- Vue.js - <https://vuejs.org/>
- React - <https://reactjs.org/>

Write a brief comparison in your report.

Again, there are insufficient marks allocated to this task, and there is no lecture support.

It is possible to attain full marks for this assignment without completing this challenge task.

Further details

Documentation

Include a written report containing:

- A list of parts of the assignment you have completed or not completed (a *statement of completion*).
- Details of specific assistance you received from people other than your lecturer or tutor, and the names of those assisting (a *statement of assistance*).
- Anything interesting or cool you'd like to draw your marker's attention to (a *statement of awesomeness*).

Assignment support

This assignment is supported by the first 10 weeks of lectures and labs. Work on the assignment should be spread over several weeks after the relevant lab has been mastered.

Submission

All files should be submitted to Moodle by the due date and time.

Marking Criteria/Rubric

Refer to the attached marking guide.

Feedback

Feedback will be supplied through Moodle. Authoritative marks will be published through fdIMarks

Academic Integrity

To submit your assessment task, you must indicate that you have read and understood, and comply with, the Federation University Australia Academic Integrity and Student Plagiarism procedures (https://policy.federation.edu.au/academic_governance/procedures/academic_integrity/ch02.php).

You must also agree that your work has not been outsourced, and is entirely your own except where work quoted is duly acknowledged. Additionally, you must agree that your work has not been submitted for assessment in any other course or program.

Marking Guide: Assignment 2

Feature	Criteria	Maximum	Obtained
Core features	Sign-up with username and password	3	
	Submit a new link with title and description	3	
	Front page / global list of links	2	
Ratings	Members can rate links	2	
	Aggregate ratings	2	
	Favourites / positively rated links page	2	
	Earning Points	2	
	Hide links	2	
Non-functional requirements	API requires authentication	2	
	Passwords use appropriate hashing	2	
	Links are displayed with submitter details	1	
	Data is stored using a database	2	
	Theme-appropriate (basic) design	2	
Report	Description of software/system architecture	1	
	Statement of completion	1	
	Statement of assistance	1	
Bonus optional task - Better than Average Ratings	Meets specification, has documentation	(+2)	
Bonus optional task - Front-end Framework	Meets specification, has documentation	(+2)	
Total:		30	